



Mobility for Bharat: Resilient Shared Energy Networks for Northeastern India

Low-cost, Flood-Resilient, & Hill-
Capable Mobility



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A Comparative Analysis of Li-
ion vs. Hydrogen Swapping
Models for Guwahati





The Challenge: The "Mobility Trilemma" in Bharat



Economic Sensitivity (The ₹300 Reality)



Target user: Daily wage driver earning ~₹500–700/day.



Constraint: Cannot afford 4–5 hours of charging downtime.



Need: "Energy Anxiety" solution—refueling must be as fast as petrol (Minutes, not hours).



Ecological Volatility (The Monsoon Factor)



Risk: Annual flash floods in Guwahati submerge roads.



Impact: Grid power often cut preemptively for 24–48 hours to prevent electrocution.



Need: Infrastructure that works without live grid power



Infrastructure Gaps (The Hill & Basin)



Terrain: 15–20° steep gradients in hill settlements (Nilachal).



Gap: Standard E-Rickshaws lack the torque to carry full loads uphill.

Operational Context: Guwahati's Unique Constraints



⚡ Topography – The Basin Effect

- ➡ City acts as a bowl; Brahmaputra River backflow prevents gravity drainage.
- ➡ Result: Chronic waterlogging requiring IP68-rated (submersible) vehicle components.

⚡ Seasonal Demand Shocks

- ➡ Ambubachi Mela (June): 2.5 Lakh+ daily footfall; massive demand for hill-climb shuttles to Kamakhya Temple.
- ➡ Magh Bihu (January): Harvest festival drives surge in heavy cargo (vegetables/livestock) from rural districts.

⚡ Grid Reliability

- ➡ Rural feeders face frequent outages during Bordoisila (pre-monsoon storms), making grid-dependent mobility risky.





Technical Vector A: Li-ion Battery Swapping (BaaS)



Technology: Lithium Iron Phosphate (LFP) Swappable Batteries



Why LFP? Superior thermal stability in humid/hot weather compared to NMC.



The Economic Winner



Running Cost: **₹1.28 – ₹1.50 per km** (vs. **₹3.21/km** for Petrol 3-wheelers).



Upfront Savings: Buying vehicle without battery reduces cost by 30–40%



Limitations



Weight Penalty: 2kWh pack weighs ~12–15kg. Carrying sufficient range (100km) reduces passenger/cargo payload capacity.



Grid Dependency: Swap stations require 24/7 grid connection; vulnerable during flood power cuts.



Technical Vector B: Hydrogen Cartridge Swapping (FCEV)



Technology: Low-Pressure Metal Hydride or Composite Cartridges.



Concept: "Bottled Fuel" – Solid-state hydrogen storage, handled like LPG cylinders.



Performance Advantage



Range: 130 km+ per fill (vs. ~80km for avg E-Rickshaw).



Refueling Time: < 2 minutes (Swap and Go).



Resilience: Passive depots require no electricity to dispense fuel, ensuring continuity during blackouts.



Cost Barrier



Current Cost: Green H₂ @ \$3.5-5/kg = ₹3.50 – ₹5.00 per km (Comparable to Petrol).

Target: National mission aims for \$1/kg by 2030 -> < ₹1.00/km



Comparative Matrix: The Right Tool for the Zone

Feature	Li-ion Swapping (City)	Hydrogen Cartridge (Hills)	Advantage Context
Running Cost	₹1.28 / km	₹3.50 / km	Li-ion wins for daily urban commute
Range	80 – 100 km	130 – 150 km	H2 wins for inter-district logistics
Hill Torque	Moderate (Voltage Sag)	High (Constant)	H2 essential for Kamakhya/Hill routes
Flood Resilience	Low (Grid Dependent)	High (Passive Infra)	H2 Critical for monsoon continuity
Payload	Low (Heavy Battery)	High (Lighter Fuel)	H2 wins for Bihu/Market cargo

This table dictates our strategy. We use Li-ion where it wins—cheap, flat city travel. We use Hydrogen where Li-ion fails—heavy cargo, steep hills, and disaster resilience

Policy Tailwinds: Why Assam? Why Now?

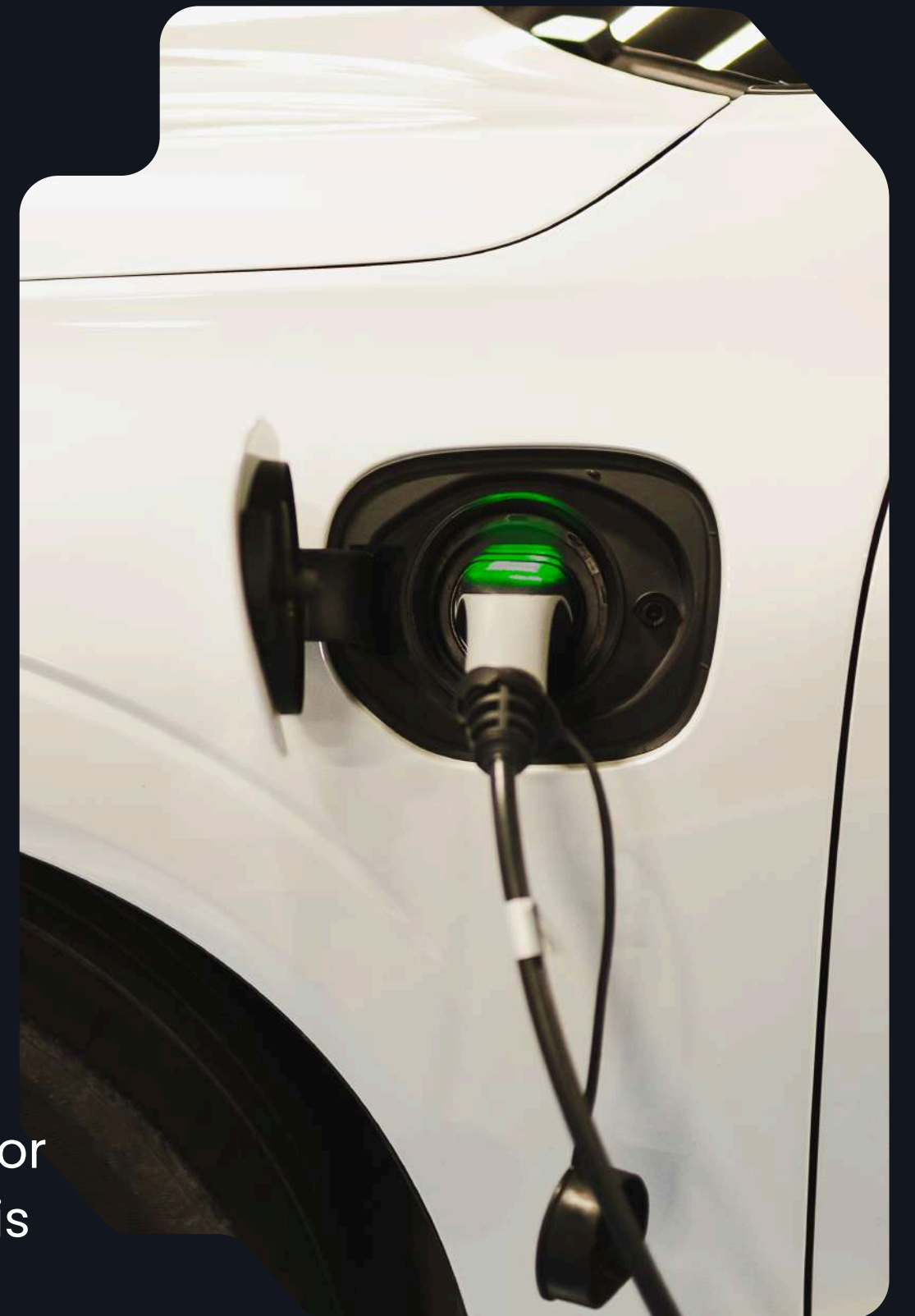
⚡ Central: PM E-DRIVE Scheme

- ➡ ₹10,900 Cr outlay; subsidies for 3.2 lakh e-3Ws reduce upfront vehicle cost.

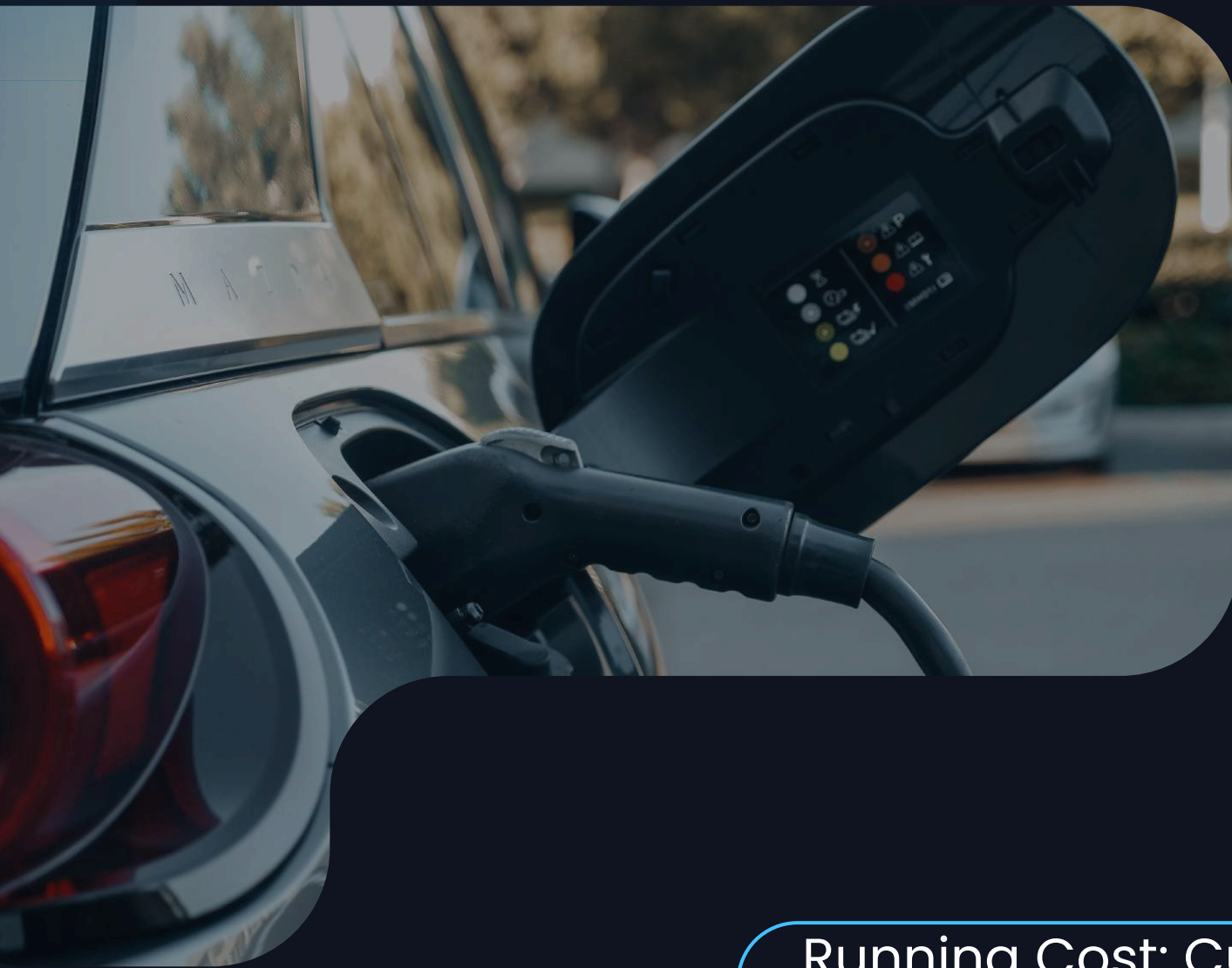
⚡ State: Assam Integrated Clean Energy Policy 2025

- ➡ H2 Production: 100% Exemption on electricity duty for Green Hydrogen production (lowers fuel cost significantly).
- ➡ Capital Subsidy: 30% subsidy on Electrolyzer and H2 infrastructure setup.
- ➡ EV Incentives: 100% Road Tax & Registration waiver for 5 years.

Assam is the perfect lab. The state policy specifically exempts electricity duty for hydrogen production. Since electricity is 70% of the cost of green hydrogen, this policy directly bridges the cost gap between Hydrogen and Li-ion



OUR SOLUTION



⚡ Zone A: Urban Core (Basin)

- ➡ Tech: Li-ion L5 E-Rickshaws with Swapping.
- ➡ Infrastructure: "Machan" (Stilt) Swap Stations elevated >4ft to survive flash floods.
- ➡ Focus: Short commutes, Last-mile connectivity to Metro/Bus stops

⚡ Zone B: Hills & Peri-Urban

- ➡ Tech: Hydrogen Cargo 3-Wheelers.
- ➡ Infrastructure: Passive "Cartridge Depots" at rural bus stands.
- ➡ Focus: Vegetable logistics (North Guwahati to Fancy Bazar), Hill transport.

Running Cost: Currently, Green Hydrogen in India costs between ₹300–₹420/kg. Assuming a 3-wheeler efficiency of 100km/kg, the running cost is ₹3.00–₹4.50 per km, which is higher than electric and comparable to petrol.

Future Parity: The National Green Hydrogen Mission targets a production cost of \$1/kg by 2030. If achieved, running costs would drop to <₹1.00/km, beating even grid electricity.



⚡ Infrastructure Economics & Driver Impact



infrastructure economics



Micro-Swap Station CAPEX: ~₹11 Lakhs (Civil + Electrical + 15 Battery inventory).



Viability: Low entry barrier allows local kirana stores to become energy franchisees.



Driver Economics



Daily OPEX: ₹165 (EV) vs ₹370 (Petrol) -> Savings: ~₹6,000/month.



Financing: "Pay-As-You-Earn" model—loan installment deducted daily from wallet recharge, removing credit score barriers.

Roadmap & Conclusion



Phase 1 (Immediate): Deploy 500 Li-ion E-Rickshaws in Guwahati Core; Retrofit 50 existing auto-rickshaws.



Phase 2 (Pilot): Launch 50 H2-Cargo Vehicles for the North Guwahati vegetable corridor utilizing NRL's pilot H2 plant.



Phase 3 (Scale): Expand Hydrogen network to Upper Assam hill districts once TCO hits $< ₹2/\text{km}$ (approx 2027-28).



Verdict: A hybrid approach is the only way to solve the "Mobility Trilemma" for Bharat's diverse geography.

